JJJ-S-701F
June 25, 1974
SUPERSEDING
Fed. Spec. JJJ-S-701E
June 22, 1970

FEDERAL SPECIFICATION

STARCH, LAUNDRY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers four types of laundry starches (see 6.1).

1.2 Classification.

1.2.1 Types and classes. The starch covered by this specification shall be of the following types and classes, as specified (see 6.2).

Type I - Thin-boiling, lubricated, noncongealing.

Class 1 - Single starch (corn or wheat).
Class 2 - Blended starches consisting of equal parts corn and wheat.

Type II - Thin-boiling, nonlubricated.

Class 1 - Blended starches consisting of equal parts corn and wheat.

Type III - Thick-boiling, lubricated.

Class 1 - Single starch (corn or wheat).

Type V - Non-boiling, lubricated, instant.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

P-S-683	-	Sour, Laundry (Fluoridated).
UU-S-48	-	Sacks, Shipping, Paper.

FSC 7930

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PPP-B-26	-	Bag, Plastic, Polyethylene.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-D-723	-	Drums, Fiber.

Federal Standards:

FED-STD-123	-	Marking for Domestic Shipment (Civil agencies).
FED-STD-191	-	Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-C-10296	-	Cloth, Cotton, Sateen, Carded.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads for 40" x 48" Pallets.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. Postal Service Manual

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Freight Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The material shall be white to cream in color and shall be substantially free from lumps, caking, or any foreign matter.

3.2 Composition and physical requirements. The starch shall be prepared from corn, or wheat, or blends of corn and wheat starch, as specified (see 1.2.1 and 6.2), and shall be in powder or granular form. A solution of the product, (prepared in accordance with directions) shall be suitable for starching clothing and other articles using accepted laundry procedures. The physical requirements of the starch shall be as specified in table I when tested as specified in 4.2.2.4.

TABLE I. Physical requirements

Characteristics	Requirement	
	Minimum	Maximum
Moisture and matter volatile at 140°C., percent, all types		13.0
Lubricants and preservatives (sum of ether and alcohol extracts), percent:		
Types I, III, and V	1.0	5.0

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TABLE I. Physical requirements (cont'd)

Characteristics	Requirement	
	Minimum	Maximum
Alkali fluidity:		
Types I and II	35 ml.	65 ml.
Type III	40 ml.	50 ml.
Type V	30 ml.	38 ml.
Congealing test (type I only)	Pass	

3.3 Uniformity of starch solution. The cooked starch solutions shall be fluid, smooth, uniform, and free from lumps when prepared in the manner specified in 4.3.6 and tested as specified in 4.2.2.4.

3.4 Starching properties (stiffness), all types. The starches, when applied to an 8-ounce cotton sateen test cloth (see 6.4), and tested as specified in 4.2.2.4, shall produce an increase in cloth stiffness (bending moment) a minimum of 8×10^{-4} inch pound.

3.5 Odor. The starch shall have no unpleasant or objectionable odor (may be mildly perfumed to impart a pleasant odor) nor shall it develop an objectionable odor when cooked up and used in the laundry.

3.6 Labeling. The following information, as applicable, shall be legibly printed on the unit container with a black water-resistant ink:

Specification
Type
Class
Name and address of manufacturer
Lot number
Directions for preparation of starch solutions, (as applicable).
Types I and II (thin-boiling) 8 ounces per gallon.
Type III (thick-boiling) 6 ounces per gallon.
Type V (instant) 4-5 ounces per gallon.

3.7 Average net content. The average net content per unit container shall be not less than specified (see 6.2).

3.8 Ease of dispersion (all types). The test starch shall be completely wetted and shall form a uniform dispersion in water when tested as specified in 4.2.2.4.

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3.9 Settling limitation (type V). After standing undisturbed for 90 minutes, the bottom starch layer in the dispersion shall occupy a volume of not less than 45 ml, when tested as specified in 4.2.2.4.

3.10 Workmanship. The starch shall be a clean, uniform product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.2.2.3. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in filled unit containers of the same type and size, containing starch of the same type and class, for the examinations in 4.2.2.1.1 and 4.2.2.1.2, in units of shipping containers for the examination in 4.2.2.2.1, or in units of pallets for the examination in 4.2.2.2.2.

4.2.2.1.1 Examination of unit containers for visual defects. The sample unit for this examination specified in table II shall be one filled unit container.

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TABLE II. Examination of unit containers

Examine	Defect
Contents	Lumpy or caked; not powdered or granular form. Not uniform. Not clean. Evidence of foreign matter.
Color	Not white to cream.
Odor	Objectionable or unpleasant.

4.2.2.1.2 Examination for average net content. The sample unit for this examination shall be one filled container. The lot shall be unacceptable if the average net content per unit container is less than specified.

4.2.2.2 Examination of preparation for delivery.

4.2.2.2.1 Examination for packaging, packing and marking. An examination shall be made to determine that packaging, packing, and markings comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit for this examination shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Marking (interior and exterior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified. Not in accordance with contract requirements.
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container. Improper closure of polyethylene bags.
Content	Number of packages or quantity per container, less than specified.

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4.2.2.2.2 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.2.2.3 Inspection levels and acceptable quality levels (AQL's) for examination. The inspection levels for determining the sample size and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL's
4.2.2.1.1	I	2.5
4.2.2.1.2	S-2	N/A
4.2.2.2.1	S-2	2.5
4.2.2.2.2	S-1	6.5

4.2.2.4 Testing of the end item. The end item shall be tested for the characteristics specified in table IV. The lot size shall be expressed in pounds of starch of the same type and class and manufactured as a single batch. The sample unit for testing shall be a 2 pound composite obtained by combining equal portions from samples selected at random throughout the lot. The frequency shall be as indicated in table III below. The composite sample shall be throughly mixed and placed in a clean, dry, sealed container with identification and date. Care shall be exercised to prevent contamination or alteration of the composite during the sampling, compositing, storage and testing. The lot shall be unacceptable if the composite fails to meet any test requirement specified.

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TABLE III. Sampling for end item testing

Lot size (pounds)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

TABLE IV. Instructions for testing of the end item

Characteristic	Specification reference: Requirement	Specification reference: Test method	Requirements applicable to: Composite	Requirements applicable to: Lot aver.	Number determinations per composite sample	Results reported as: Pass or fail 2/	Results reported as: Numerically to nearest 3/
Composition	3.2	1/					
Moisture and matter volatile at 140°C all types	Table I	4.3.3	X		2 (average)		0.1 perce
Presence of lubricants, preservatives, etc. (expressed as the sum of the ether and alcohol extracts): Types I, III, and V	Table I	4.3.4 & 4.3.5	X		2 (average)		0.1 percent
Alkali fluidity of starch solution: Types I, II, III and V	Table I	4.3.6, 4.3.6.1, 4.3.6.2, 4.3.6.3, 4.3.6.4, and 4.3.6.5			2		4 centipoises

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TABLE III. Sampling for end item testing (cont'd)

Characteristic	Specification reference		Requirements applicable to		Number determinations per composite sample	Results reported as	
	Requirement	Test method	Composite	Lot aver.		Pass or fail 2/	Numerically to nearest 3/
Congealing test (type I only)	Table I	4.3.8	X		1	X	
Starching test (all types)	3.4	4.3.9					
Ease of dispersion	3.8	4.3.11	X		1	X	
Settling limitation (type V)	3.9	4.3.12	X		1	X	

1/ Certificate of compliance will be furnished to the contracting officer for composition requirements.

2/ If failure is indicated, report either descriptive or numerical point of failure, as applicable.

3/ Test reports shall include all values on which results are based.

4.3 Test methods.

4.3.1 Test materials.

Petroleum ether - reagent grade, conforming to ACS standards.
Ethyl alcohol - U.S.P. grade, 95 percent.
Sodium Hydroxide - 0.375 N.
Cloth - 8-ounce sateen cotton cloth conforming to MIL-C-10296, class 1. The cloth shall be dry before use.

4.3.2 Test equipment.

1. Oven, air circulating.
2. Soxhlet extraction apparatus.

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3. Balance, analytical, accurate to 0.1 mg.
4. Balance, 5000 gram capacity.
5. Bottle, weighing, with tapered ground stopper, 6-8 cm diameter, 2-4 cm depth.
6. Electric hot plate.
7. Graduated cylinders - glass stoppered 250 ml capacity, and widemouth-type, 500, and 1000 ml capacity.
8. Thermometers - 0°-150°C or 32°-300°F.
9. Beaker, stainless steel, 1500 ml capacity.
10. Cover for 1500 ml beaker - Plexiglass and cut to accomodate triangular stirrer, steam line and thermometer.
11. Triangular stirring rod - Made from a 1/4-inch diameter metal rod with the base of the triangle 3-inches in length.
12. Variable speed motor with voltage regulator (Variac).
13. Glass rods - 5 mm OD x 200 mm length.
14. Live steam injection system consisting of:
    a. low pressure steam - 15 psi
    b. condensate trap
    c. shut-off valve
    d. rubber pressure tubing
    e. stainless steel tube - 3/8-inch OD and 1/4-inch ID, 8-inch length
15. Beaker, Pyrex - 150 ml and 400 ml capacity.
16. Constant temperature water bath maintained at 40° ± 1°C.
17. Fluidity funnel.
18. Stop watch.
19. Tumble jar equipment - Described in Dynamic Absorption Test, Method 5500.1 of FED-STD-191.
20. Pressing equipment - A flat-bed press capable of pressing a specimen measuring 18 x 24 inches and equipped with a temperature control to maintain the temperature between 275° and 300°F.
21. Conditioning room - A room where the temperature is maintained at 70° ± 2°F and the relative humidity at 65 ± 2% RH.
22. A die and other equipment required to strike out fabric specimens measuring 6 x 1 inches (see 6.3).
23. Stiffness tester - Described in Method 5206.1 of FED-STD-191.
24. Lightning stirrer.

4.3.3 Moisture and matter volatile at 140°C, all types. Weigh 5 grams of the starch sample to the nearest milligram in a tared weighing bottle (provided with a ground stopper) about 6-8 cm in diameter and 2-4 cm in depth. Place in an air oven at a temperature of 140° ± 2°C and heat for 2-1/2 hours. Remove the bottle and contents, cool in a desiccator for 30 minutes, and weigh. Calculate percent loss in weight.

% Moisture and matter volatile =

$$\frac{\text{Original weight of sample - Weight of oven-dried sample} \times 100}{\text{Original weight of sample}}$$

A moisture balance may also be used for moisture determination.

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4.3.4 Ether extract, types I, III, and V. Weigh 5 grams of starch sample to the nearest milligram and transfer quantitatively to a Soxhlet extraction thimble. Loosely plug the opening of the thimble with a wad of extracted cotton. Introduce the thimble into a Soxhlet extraction tube. Fill a tared oil-extraction flask about three-quarters full with petroleum ether. Assemble the extraction apparatus using ground glass connections or cork stoppers. Extract for 4 hours. Disconnect the apparatus and drain the ether from extraction tube back into the flask. Remove the thimble and contents from the extraction tube and set aside for later analysis Recover the ether by distilling it into the extraction tube. Boil off the last trace of the ether and place flask in an air oven at 105° ± 2°C until constant weight is obtained. The weight of the residue in the flask, in grams, times 20, equals the percent ether extract.

4.3.5 Alcohol extract, types I, III, and V. Replace the thimble and contents after the petroleum ether extraction in the Soxhlet extraction tube. Fill a tared oil-extraction flask about three-quarters full with 95 percent ethyl alcohol and reassemble the apparatus. Extract for 4 hours; disconnect the apparatus, and evaporate alcohol from the flask on a steam bath. Dry to constant weight at 105° ± 2°C in an air oven. The weight of the residue in the flask, in grams, times 20, equals the percent alcohol extract.

4.3.6 Alkali fluidity. The alkali fluidity of the starch solutions shall be determined by using the following procedure.

4.3.6.1 Standardization of funnel. Select a funnel having an orifice such that 100 mls of distilled water shall pass through it in the range of 55 to 75 seconds. Measure 100 mls of distilled water at 70-75°F. Pour into the funnel with a plunger in place. Lift the plunger and record the time in seconds for the 100 mls of water to pass through.

4.3.6.2 Test procedure.

4.3.6.3 Types I and II (dry starch). Make into a slurry, 5.0 grams of oven-dried starch (1 hour at 100°C), 10 mls of distilled water, 90 mls of 0.375 N. NaOH and stir with a Lightning stirrer for exactly 3 minutes. The slurry is then poured into the funnel which has been plugged from the top with the plunger. The funnel then is unplugged. Collect in a 100 ml graduated cylinder and record the volume of the slurry which passes through the funnel in the same time which was required for the 100 mls of distilled water to pass. This figure represents the fluidity of the sample (the higher the fluidity unit, the thinner or more fluid the product).

4.3.6.4 Types III and V (dry starch). Follow the same procedure as described in 4.3.6.3, except that the amount of oven-dried starch used shall be 2.0 grams.

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4.3.6.5 Starch slurry. If it is desired to measure the alkali fluidity of a starch solution during manufacture, the following procedure may be used:

Example: For a 2.0 gm sample,

a 21° Baume solution = 3.694 lbs./gal.

$$\frac{3.694 \times 454 \times 1000}{3785} = 442 \text{ gms/1000 mls.}$$

$$\frac{442}{2.0} = \frac{1000}{X} = 4.5 \text{ mls of slurry (equal to 2.0 gms of dry starch)}$$

To the slurry, add enough 0.375 N NaOH solution to make 100 mls total. Place on stirrer and stir for exactly 3 minutes. Place plunger* in position in funnel and pour the alkali-starch mixture into the funnel. Collect in a 100 ml graduated cylinder the volume of starch solution which passes through the funnel in the same time as is required for the 100 ml of distilled water at 70-75°F to pass. The number of mls collected represents the alkali fluidity of the starch.

* The plunger may be fashioned from a glass rod of suitable size with a rubber cap from an eyedropper inserted on one end.

4.3.7 Preparation of starch solutions.

4.3.7.1 Types I and II starches. Solutions of types I and II starches (6.0 percent by weight anhydrous starch, approximately 8.0 oz. per gallon) shall be prepared and determined as follows:

To a tared 1500 ml stainless steel beaker with plexiglass cover, add 700 ml distilled water at room temperature. Weigh out 60.0 g (anhydrous basis) starch and add to the beaker containing 700 ml distilled water and slurry mechanically for 10 minutes using a variable speed stirrer equipped with a triangular shaped stirring rod (see 4.3.2 item 11). The stirring rod shall be just off the bottom of the beaker. The speed of motor shall be such that a homogeneous slurry is maintained without loss of materials. Insert a thermometer in beaker to measure temperature of starch paste during the cooking operation. Introduce a live steam injection line below the surface of the starch slurry to within 3/4-inch of the bottom of the beaker. Adjust steam addition to produce a boiling paste in 5 to 10 minutes. Carefully control flow of steam so as not to cause a loss of material by splashing. The slurry should continue to be stirred mechanically throughout the cooking period increasing or decreasing the speed of the motor as needed to keep the paste moving or to avoid excessive bubbling and loss of materials. When necessary, rotate the beaker as the slurry begins to paste, so that the starch cooks evenly. Upon obtaining a

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boiling paste (210°-212°F), continue to boil for 20 minutes. Remove steam and mechanical stirrer, adjust the weight of the cooked starch paste to 1000 g with distilled water; stir the cooked paste with a glass rod during the addition of the water to avoid formation of lumps. Into a tared 150 ml beaker, pour 100 gms of the type I hot starch paste, cover top of beaker with aluminum foil to prevent evaporation loss and "skinning", and set aside for the congealing test (see 4.3.8). Place beaker containing the remaining 900 g of starch paste in constant temperature bath, cover, and allow paste to cool down to the temperature of the bath, 102°-106°F (39°-41°C), stirring occasionally to keep scum and lumps from forming.

4.3.7.2 Type III starch. A solution of type III starch (3.75 percent by weight, anhydrous starch, approximately 5.0 oz per gallon), shall be prepared and the viscosity determined as follows:

Weigh accurately 37.5 g (anhydrous basis) of the starch sample. Proceed according to the method specified in 4.3.7.1.

4.3.8 Congealing test (type I starch). Congealing test shall be determined on the starch paste set aside as specified under 4.3.7. Allow the sample of starch paste to cool, undisturbed, to a temperature of 68°-77°F (20° to 25°C). After standing at that temperature for 3 to 4 hours, the solution shall not congeal to the extent that it cannot be poured from the beaker.

4.3.9 Starching test - types I, II, and III.

4.3.9.1 Preparation of test specimen. Three (3) pounds of the "standard" test fabric (8-ounce cotton sateen cloth, MIL-C-10296, class 1, cut into 1 yd pieces), shall be laundered in a washwheel according to Method 5550.2 of FED-STD-191. Specimens measuring 10 inches in the warpwise direction and 16 inches in the fillingwise direction shall be cut from the fabric. Do not cut specimens closer than 3 inches from the selvedge.

4.3.9.2 Preparation of cooked starch solution. With types I and II starches, solutions of 6.0 percent by weight of anhydrous starch shall be prepared in accordance with the procedure outlined in 4.3.7.1. These freshly-prepared starch solutions shall be applied to the test fabric as outlined in a, b, and c of 4.3.9.3. In the case of types III and V starches, a solution of 3.75% by weight of anhydrous starch shall be prepared.

4.3.9.3 Application of starch to test fabric, types I, II, and III.

a. Dissolve 2.0 g of laundry sour, sodium silico fluoride and sodium acid fluoride (P-S-683, type 1) in 1210 mls of distilled water at 70°-80°F and pour the solution into a tumble jar. Weigh 227 $\pm$ 2 g test fabric specimens. Add all of these specimens (227 g) to the jar and rotate for 10 minutes.

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b. Weigh 152 ± 1 g of cooked starch solution (39°-41°C) in a 400 ml beaker. Dilute this starch with about 200 ml of the fluoride sour solution removed from the tumble jar and mix throughly. Pour the diluted starch solution into the tumble jar containing the remainder of the sour solution and rotate the jar for 20 minutes.

c. Previously adjust the wringer rolls to permit a wet pick-up in the specimens of approximately 115% (do not use the test specimens for this purpose), remove the test fabric from the jar, hand squeeze to remove excess starch solution and hand smooth to remove wrinkles and creases. Pass the specimens one at a time between the squeeze rolls. After weighing, immediately wrap the squeezed test fabric specimens in aluminum foil to retard the loss of moisture by evaporation. Press the specimens dry within 2 hours.

$$\% \text{ wet pick-up} = \frac{\text{Weight of wet specimens - Weight of orig. dry specimens x 100}}{\text{Orig. weight of dry specimens}}$$

4.3.9.4 Pressing of test fabric. Press the starched test fabric specimens using a flat-bed press. Operate the press at a temperature between 275° and 300°F. Press the test specimen between two 18" x 24" clean white cotton percale press cloths, placing one of the press cloths on the bed of the press and the other on the top of the test specimens (change press cloths frequently, especially if testing different starches or when testing a control specimen). Make certain no wrinkles are present and that any loose threads are removed so as not to be pressed into the specimens. Lower the "head" of the press down onto the specimens for 30 seconds and then release the pressure. Use no open steam on the press. Check the specimens for wrinkles and creases (smooth out if present) and lower the head for another 30 seconds. Repeat this procedure until the specimens are dry to the touch. Remove specimens with care, to prevent wrinkling or creasing, and place, single thickness, on a flat, horizontal screen or rack for a minimum of 24 hours, to condition, in an atmosphere maintained at 70° ± 2°F and 65 ± 2 percent relative humidity.

4.3.9.5 Fabric stiffness determinations. Select from the conditioned specimens two which are uniform in appearance and free from wrinkles and creases. Cut from each, test specimens measuring 6 x 1 inches, a minimum of 5 in the warpwise direction and 5 in the fillingwise. Determine the flex stiffness of each specimen in accordance with Method 5206.1 of FED-STD-191 and express the results to the nearest 1 x 10^{-4} inch pounds. The sateen "face" of the test specimen shall be placed down on the apparatus when determining "stiffness".

Increase in stiffness = S-O

Where:

S = Stiffness of starched sateen cloth, inch pounds.
O = Stiffness of original unlaundered, unstarched sateen, inch pounds.

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4.3.10 Application of starch to test fabric - type V. This starch shall be tested by the same procedure outlined above, except that a dispersion consisting of 5.7 g of dry starch stirred into 146.3 g of distilled water (70°-80°F) (3.75 percent), shall be used in the place of 152 g of cooled starch solution.

4.3.11 Ease of dispersion (all types of starch). Measure 250 ml of distilled water (125°-130°F) into a 400 ml beaker. Add 18.0 ± 0.1 g of the starch to the water, using a spatula having a 5 inch blade, and stir continuously, at a moderate rate for 5 minutes. Note any floating material or undispersed starch.

4.3.12 Settling limitation (type V starch). The dispersion prepared as outlined in 4.3.11 is poured into a glass-stoppered 250 ml graduated glass cylinder having a minimum length of 9-1/2" between the 0 and the 250 ml markings and a minimum of 1-1/2" between the 250 ml marking and the bottom of the glass stopper of the cylinder. Stopper and mix the contents by inversion for one minute at the rate of 60 inversions per minute (one inversion equals one complete cycle of movement; i.e., backward and forward). The mixing shall be carried out by holding the elbows near the hips and moving the cylinder in an arc across the body, back and forth, at the prescribed rate. The sample then is set aside for 90 minutes at which time the volume of the bottom layer, if any, is noted.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Starch shall be furnished in 1-pound, 50-pound and 265-pound quantities as hereinafter specified (see 6.2).

5.1.1.1 One-pound quantity. One pound of starch of one type and class only shall be packaged in a snug-fitting paperboard box conforming to variety 1, style I, type A, class a or b of PPP-B-566. Each box shall be securely closed in accordance with the appendix of the box specification.

5.1.1.2 Fifty-pound and 265-pound quantities. Packaging shall not be required.

5.1.2 Level C, commercial packaging. Starch shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Starch shall be furnished in 1-pound, 50-pound and 265-pound quantities as hereinafter specified (see 6.2).

5.2.1.1 One-pound quantities. Forty 1-pound packages of starch of one type and class only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

5.2.1.2 Fifty-pound quantities. Fifty pounds of starch of one type and class only shall be packed in a multiwall paper shipping sack conforming to type I, II, or IV, construction number 8 or 8X of UU-S-48, with top and bottom ends dipcoat sealed as specified therein. Closure shall be in accordance with the sack specification. Alternatively, fifty pounds of starch of one type and class only shall be packed in a fiberboard drum fabricated, assembled and closed in accordance with the applicable requirements for type III, grade A, class 1 of PPP-D-723. Each drum shall be lined with a minimum 0.002 inch thick polyethylene bag conforming to type I, grade A, style 1 or 2 of PPP-B-26. The plastic bag shall be closed in a manner to prevent sifting of the product through the closure.

5.2.1.3 Two hundred and sixty-five-pound quantities. Two hundred and sixty-five pounds of starch of one type and class only shall be packed in a fiberboard drum, fabricated, assembled and closed in accordance with the applicable requirements for type III, grade A, class 1 of PPP-D-723. Each drum shall be lined with a minimum 0.002 inch thick polyethylene bag conforming to type I, grade A, style 1 or 2 of PPP-B-26. The plastic bag shall be closed in a manner to prevent sifting of the product through the closure.

5.2.2 Level B. Starch shall be furnished in 1-pound, 50-pound and 265-pound quantities as hereinafter specified (see 6.2).

5.2.2.1 One-pound quantities. Forty 1-pound packages of starch of one type and class only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

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5.2.2.2 Fifty-pound quantities. Fifty pounds of starch of one type and class only shall be packed in a multiwall paper shipping sack conforming to type I, II, or IV, construction number 4 or 4X of UU-S-48. Closure shall be in accordance with the sack specification. Alternatively, fifty pounds of starch of one type and class only shall be packed in a fiberboard drum, fabricated, assembled and closed in accordance with the applicable requirements for type I, grade A, class 1 of PPP-D-723. Each drum shall be lined with a minimum 0.002 inch thick polyethylene bag conforming to type I, grade A, style 1 or 2 of PPP-B-26. The plastic bag shall be closed in a manner to prevent sifting of the product through the closure.

5.2.2.3 Two hundred and sixty-five-pound quantities. Two hundred and sixty-five pounds of starch of one type and class only shall be packed in a fiberboard drum, fabricated, assembled and closed in accordance with the applicable requirements for type I, grade A, class 1 of PPP-D-723. Each drum shall be lined with a minimum 0.002 inch thick polyethylene bag conforming to type I, grade A, style 1 or 2 of PPP-B-26. The plastic bag shall be closed in a manner to prevent sifting of the product through the closure.

5.2.3 Level C, commercial packing. Starch, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable.

5.3 Palletization. When specified (see 6.2), starch, of one type and class only, packed in fiberboard shipping containers, fiberboard drums or multiwall sacks as specified in 5.2 shall be palletized in accordance with load type I, XIII or XV, as applicable of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course when applicable. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil agencies. In addition to labeling specified in 3.6, and any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

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5.4.2 Military requirements. In addition to labeling specified in 3.6, and any special marking required by the contract or order, interior packages, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The laundry starches covered by this specification are intended for institutional, military, and commercial laundry use where it is desirable to impart body and improve the "handle" and appearance of the laundered fabrics.

6.1.1 Lubricated starches. Lubricants such as waxes and sulfonated oils are added to starches during their manufacture to improve the ironing characteristics, pliability, and appearance (luster) of the finished starched fabrics.

6.1.2 Thin-boiling, lubricated, noncongealing starches (type I). Cooked solutions prepared with this type starch are liquids when allowed to cool at room temperature. Type I starches will remain fluid and usable at this temperature. If not completely used on the day prepared, they may be used on the following day.

6.1.3 Thick-boiling starches (type III). These starches impart greater stiffness to fabrics in comparison to thin-boiling starches of the same concentration and are preferred when "boardy" finishes are desired.

6.1.4 Instant starch (type V). This starch may be used without cooking. It disperses easily in cool or luke warm water with no balling or agglomerates visible after 2 minutes of normal agitation.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Quantity desired (see 3.7, 5.1.1, 5.2.1 and 5.2.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1.1).
(f) When palletization is required (see 5.3).

6.3 Steel die. Steel die may be obtained from Boston Cutting Die Company, 50 Freeport Street, Boston, MA.

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6.4 Standard reference fabric. The standard reference fabric for testing purposes may be obtained from the U.S. Army Natick Laboratories, Natick, MA 01760.

6.5 Type IV deleted; no requirement.

Custodians:

Army - GL
Air Force - 84

Review activity:

Army - MD

User activity:

Navy - MC

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS
HEW-NIH

Project No. 7930-0301

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 40 cents each.

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DOCUMENT IDENTIFIER AND TITLE

JJJ-S-701F STARCH, LAUNDRY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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